

Dynamic Set-Theoretic Normalization of Provenance-Bearing Hyper-Relational Frames

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Abstract

We develop a deterministic semantics for normalizing claims and event frames across a mutable document corpus. The extraction front end may propose several frame interpretations for each claim occurrence, and coreference is not forced to one early clustering. Instead, every exact interpretation and strict coreference partition is a candidate normalization hypothesis. Constraints denote property sets of hypotheses, and the solver is literally set theoretic estimation (STE): it repeatedly intersects its current state with property sets and records the exact set reduction. The resulting feasible family supports a possible-worlds semantics for identity. Coreference holding in every feasible hypothesis defines a safe quotient; disagreement between feasible merge and split hypotheses defines an explicit uncertainty overlay $\mathcal{U}$.

We prove document-insertion antitonicity, deletion expansion, insertion order independence, a delta decomposition by newly activated provenance constraints, identity and composition laws for feasible-world restriction, equivalence of must-coreference, and exact status separation into inconsistent, must, cannot, and uncertain cases. A finite countermodel proves that possible coreference and the uncertainty overlay need not be transitive even though every exact hypothesis is a partition. We distinguish the safe must-quotient from a possible-coreference ambiguity envelope and prove that insertion can strictly split the latter by eliminating a merge world. A two-claim Toyota/BMW model gives a constructive witness and an exact one-candidate reduction. We prove existence of finite minimal document supports for must- and cannot-coreference conclusions. We then resolve six initial conjectures under explicit, necessary qualifications: partial variable-universe restriction; gluing by a cross-constraint obstruction with at most one extension; a finite exact solver with tight 2^N state-space bound; audited typed-relation lifting; exact marginal commutation for dependent truth fibers; and occurrence-set identity with relational lineage. The accompanying Lean 4 development names every stated formal result.

1 Introduction

Cross-document event coreference is usually presented as clustering: a learned model scores pairs or clusters, and an agglomerative or graph algorithm selects one partition. This is an effective predictive architecture, but it is an awkward database semantics. A database receives and retracts documents. A new source can distinguish events previously thought identical; deletion can remove the only evidence that forced a merge or split. Anaphora and omitted arguments can remain genuinely unresolved. A destructive union operation has neither the provenance nor the inverse required by this setting.

Event identity itself is richer than lexical similarity. Work on quasi-identity and dense event annotation distinguishes strict identity from subevent, near-identity, and related-event phenomena [5, 7, 8]. Recent resources make decontextualization, difficult identity boundaries, and framing divergence increasingly explicit [9, 10]. The correct deterministic reaction is not to pretend every

ambiguity has a unique answer. It is to retain the admissible answers and state exactly what follows in all, some, or none of them.

This paper formalizes that reaction. We assume a high-recall extraction front end. Each immutable claim occurrence has source provenance and a nonempty set of candidate hyper-relational frames. An exact normalization hypothesis chooses one candidate per claim and supplies an equivalence relation of strict frame identity. Typed nonidentity links—subevent, causation, temporal relation, repeated instance, framing divergence—are conceptually separate and do not participate in the identity quotient.

The contributions are:

1. an STE solver semantics in which property-set application and exact reduction are the complete operational core;
2. a provenance-indexed corpus semantics supporting insertion and deletion without destructive clustering;
3. a must-coreference quotient and a nontransitive uncertainty overlay;
4. a finite countermodel ruling out possible-coreference clustering;
5. a mechanized insertion-induced ambiguity-envelope split, including a finite cardinal bound and an exact Toyota/BMW witness;
6. existence of minimal finite document certificates for settled links;
7. contravariant feasible-world restriction and covariant canonical-frame maps, exposing the fundamental insertion/deletion asymmetry;
8. a dependent lift of downstream truth estimation over all feasible normalizations.

2 The normalization universe

Fix types of documents, claim occurrences, structured frames, exact normalization hypotheses, and constraints:

$$D, \quad C, \quad F, \quad \Omega, \quad K.$$

The Lean structure implementing these data is `STE.DynamicFrame.Model`.

Each claim c has an immutable source document $\text{owner}(c)$ and a nonempty candidate set $\Phi(c) \subseteq F$. Each exact hypothesis $\omega \in \Omega$ chooses

$$\text{interpret}_\omega(c) \in \Phi(c)$$

and supplies a strict-coreference relation $\sim_\omega$ on claims. The model requires $\sim_\omega$ to be an equivalence relation. Thus every exact hypothesis is an ordinary, logically valid partition. Ambiguity is represented between hypotheses, not by allowing one hypothesis to contain an invalid overlapping “partition.”

Every constraint $k \in K$ has two denotations. First, its provenance support is a set of documents

$$\text{supp}(k) \subseteq D.$$

Second, it denotes an STE property set

$$S_k = \{\omega \in \Omega \mid \omega \text{ satisfies } k\}.$$

For an active corpus $D \subseteq \mathcal{D}$, the active constraints and feasible normalizations are

$$K(D) = \{k \mid \text{supp}(k) \subseteq D\}, \quad (1)$$

$$\mathcal{N}(D) = \bigcap_{k \in K(D)} S_k. \quad (2)$$

Empty-support logical or ontological constraints are active in every corpus. A constraint derived jointly from several documents is active only while every document in its support remains active.

This fixed ambient universe is deliberate. Documents, claims, constraints, and hypotheses are immutable mathematical objects. Database mutation changes an activation set. A later variable-universe theory may garbage-collect or forget inactive claims through explicit restriction maps, but semantic retraction does not require erasing the objects that explain prior results.

3 The solver is STE

The phrase “constraint solver” can suggest an additional optimization or clustering semantics. There is none here. The solver state is a set $X \subseteq \Omega$. Applying k is literal STE property-set application:

$$\text{apply}(X, k) = X \cap S_k. \quad (3)$$

The exact reduction caused by k is

$$\text{red}(X, k) = X \setminus (X \cap S_k) = \{\omega \in X \mid \omega \notin S_k\}. \quad (4)$$

These are `Model.applyConstraint` and `Model.solverReduction`.

Proposition 3.1 (One-step STE laws). *Property application is contractive and idempotent; any two property applications commute. Moreover,*

$$\text{apply}(X, k) \cup \text{red}(X, k) = X,$$

and $\text{red}(X, k) = \emptyset$ exactly when $X \subseteq S_k$.

Lean: `applyConstraint_subset`, `applyConstraint_idempotent`, `applyConstraint_comm`, `apply_union_reduction`, and `solverReduction_eq_empty_iff`.

For a finite schedule $[k_1, \dots, k_n]$, define a run by repeated application, starting from X , and define solving to start from Ω .

Theorem 3.2 (Operational–denotational correspondence). *For every finite list L ,*

$$\omega \in \text{run}(L, X) \iff \omega \in X \wedge \forall k \in L, \omega \in S_k, \quad (5)$$

$$\text{solve}(L) = \bigcap_{k \in L} S_k. \quad (6)$$

Consequently, schedules with the same constraint membership have equal output; in particular, permutations and duplicates do not matter.

Lean: `mem_run`, `run_eq_inter_partial`, `solve_eq_partialFeasibilitySet`, `solve_congr`, and `solve_perm`.

If a list enumerates exactly $K(D)$, the final operational state equals $\mathcal{N}(D)$ (`solve_eq_feasible`). Thus a SAT, SMT, ASP, BDD, or custom partition engine is correct only insofar as

it implements this meet and its queries. Such technologies are representations and acceleration strategies, not alternate semantics.

Retraction follows the same point. The system does not attempt to invert an old merge. It changes $K(D)$ using provenance and evaluates the new meet. An incremental implementation may reuse caches or truth-maintenance labels, but its extensional correctness criterion is equality with the fresh STE meet. This connects directly to truth-maintenance systems [2, 3], provenance [4], and incremental view maintenance [1].

4 Dynamic corpus laws

Let $D \subseteq E$. Every constraint active in D is active in E , hence

Theorem 4.1 (Information antitonicity).

$$D \subseteq E \implies \mathcal{N}(E) \subseteq \mathcal{N}(D).$$

Lean: `Model.feasible_antitone`. Read forward, document addition removes hypotheses. Read backward, removal can restore hypotheses. The latter is `feasible_subset_after_remove`.

The elementary database laws are exact:

duplicate insertion is idempotent (`feasible_insert_idempotent`);

insertion order is irrelevant (`feasible_insert_comm`);

if $d \notin D$, adding and then removing d restores precisely $\mathcal{N}(D)$ (`feasible_insert_then_remove`).

Define the constraint delta

$$\Delta K(D, E) = K(E) \setminus K(D).$$

Then addition has an exact incremental decomposition.

Theorem 4.2 (Delta decomposition). *If $D \subseteq E$, then*

$$\omega \in \mathcal{N}(E) \iff \omega \in \mathcal{N}(D) \wedge \forall k \in \Delta K(D, E), \omega \in S_k.$$

Lean: `mem_feasible_extension`. If the delta is empty, the feasible sets are equal (`feasible_eq_of_no_new_constraints`).

Constraint activation interacts asymmetrically with union and intersection:

$$K(D \cap E) = K(D) \cap K(E), \quad K(D) \cup K(E) \subseteq K(D \cup E).$$

The second inclusion can be strict because $D \cup E$ may activate a constraint whose support draws documents from both sides. Lean: `activeConstraints_inter` and `activeConstraints_union_subset`. This is the first genuinely cross-document effect in the theory.

5 Must, may, cannot, and uncertainty

For $c, d \in \mathbb{C}$ and a consistent corpus, define

$$\text{Must}_D(c, d) \iff \forall \omega \in \mathcal{N}(D), c \sim_\omega d, \quad (7)$$

$$\text{May}_D(c, d) \iff \exists \omega \in \mathcal{N}(D), c \sim_\omega d, \quad (8)$$

$$\text{Cannot}_D(c, d) \iff \forall \omega \in \mathcal{N}(D), c \not\sim_\omega d, \quad (9)$$

$$\text{U}_D(c, d) \iff (\exists \omega \in \mathcal{N}(D), c \sim_\omega d) \wedge (\exists \omega \in \mathcal{N}(D), c \not\sim_\omega d). \quad (10)$$

This is the certain/possible distinction of incomplete-database possible worlds [6] applied to normalization partitions.

Consistency must be checked first. If $\mathcal{N}(D) = \emptyset$, both universal statements are vacuously true. The implemented status type therefore has four constructors:

inconsistent, must, cannot, uncertain.

Lean: `Model.CorefStatus` and `Model.corefStatus`, with one correctness theorem for each constructor.

Theorem 5.1 (Safe quotient). *Must_D is an equivalence relation. Therefore the deterministic normalized frame collection*

$$F'_D = \mathbb{C}(D)/\text{Must}_D$$

is well defined on active claims.

Proof. Reflexivity, symmetry, and transitivity hold in every exact relation $\sim_\omega$ and are preserved by intersection over $\omega \in \mathcal{N}(D)$. $\square$

Lean: `mustSame_equivalence`, `activeMustSetoid`, and `CanonicalFrame`.

Possible coreference is reflexive on a consistent corpus and symmetric, but no general transitivity theorem is available. U_D is symmetric and irreflexive. Must and cannot are each disjoint from uncertainty. Lean: `maySame_refl`, `maySame_symm`, `uncertain_symm`, `uncertain_irrefl`, `must_not_uncertain`, and `cannot_not_uncertain`.

5.1 A finite countermodel

Let the mentions be a, b, c and let two exact worlds be feasible. The left world has blocks $\{a, b\}, \{c\}$; the right world has blocks $\{a\}, \{b, c\}$. Each world is an ordinary partition. Nevertheless,

$$\text{May}(a, b), \quad \text{May}(b, c), \quad \neg \text{May}(a, c).$$

Moreover (a, b) and (b, c) are uncertain while (a, c) is cannot-corefer.

Theorem 5.2 (Nontransitivity). *Neither May nor U is transitive in general.*

The model and proofs are fully finite and constructive in `Ste.DynamicFrame.Counterexample`; the theorem names are `maySame_not_transitive` and `uncertain_not_transitive`.

This establishes a design constraint. U is an overlay between must-classes, not a special kind of class. Taking connected components of possible or uncertain edges invents identifications not licensed in any exact world.

5.2 Insertion can split an ambiguity envelope

The preceding warning does not mean that an existing database-facing frame can never split. It means that we must say which object bears the word “frame.” Define the possible-coreference envelope anchored at c by

$$A_D(c) = \{d \in \mathbf{C} : \text{May}_D(c, d)\}.$$

This is a presentation of unresolved alternatives, not an equivalence class. For $D \subseteq E$, feasibility antitonicity gives

$$A_E(c) \subseteq A_D(c).$$

The inclusion can be strict. Say that insertion *resolves* c, d *apart* when

$$\text{U}_D(c, d) \quad \text{and} \quad \text{Cannot}_E(c, d).$$

Theorem 5.3 (Insertion-induced envelope splitting). *If $D \subseteq E$ resolves c, d apart, then*

$$A_E(c) \subseteq A_D(c), \quad d \in A_D(c), \quad d \notin A_E(c).$$

If the claim universe is finite, then

$$|A_E(c)| < |A_D(c)| \leq |\mathbf{C}|.$$

Thus the exact envelope reduction $|A_D(c)| - |A_E(c)|$ is positive and is at most $|\mathbf{C}|$ without further assumptions.

Proof. The old uncertainty supplies a feasible world merging c and d , hence $d \in A_D(c)$. The new cannot-link is equivalent to absence of every feasible merge world, hence $d \notin A_E(c)$. Antitonicity of **May** supplies the subset. For finite sets, a proper subset has strictly smaller cardinality; both envelopes are subsets of the ambient claim universe. $\square$

Lean: `Model.mayEnvelope_antitone` and `Model.mayEnvelope_split_witness`.

The finite theorem names are

```
Model.finiteMayEnvelope_card_lt_of_resolvesApart
Model.envelopeReduction_le_card.
```

The smallest informative witness contains two old descriptions and two exact worlds. Initially c is “the red car” and d is “the burgundy car.” One world treats them as one vehicle and the other as two, so

$$\mathcal{N}(D) = \{\{\{c, d\}\}, \{\{c\}, \{d\}\}\}.$$

A later document identifies the red description as Toyota and the burgundy description as BMW. Under the audited property that one vehicle cannot have both manufacturers, STE rejects the one-vehicle world:

$$\mathcal{N}(E) = \{\{\{c\}, \{d\}\}\}.$$

Consequently $A_D(c) = \{c, d\}$, $A_E(c) = \{c\}$, and the exact reduction is 1. This construction is mechanized in module `Ste.DynamicFrame.AmbiguitySplit`, namespace `VehicleSplit`; the principal results are `toyota_bmw_envelope_split` and `toyota_bmw_split_into_two_singletons`, with the exact reduction certified by `toyota_bmw_reduction_eq_one`.

There is no contradiction with monotonicity of the must-quotient. If $\text{Must}_D(c, d)$ really held, constraint intersection could not make it false: every later world was already an earlier world. The split occurs because the old database-facing object was an ambiguity envelope containing a *possible* link, not a forced identity class. The authoritative state is therefore the feasible family of exact partitions. A single partition plus pairwise **U** marks is only a summary and can lose correlations among alternatives.

6 Restriction, quotient maps, and retraction asymmetry

On the fixed ambient hypothesis universe, inclusion $D \subseteq E$ induces

$$\rho_{E,D} : \mathcal{N}(E) \longrightarrow \mathcal{N}(D), \quad \rho_{E,D}(\omega) = \omega.$$

This is a genuine restriction map: it forgets the additional requirements, not the underlying hypothesis.

Theorem 6.1 (Restriction laws). *Restriction by identity is identity, and for $D \subseteq E \subseteq F$,*

$$\rho_{E,D} \circ \rho_{F,E} = \rho_{F,D}.$$

A point of $\mathcal{N}(D)$ lies in the image of $\rho_{E,D}$ exactly when the same ambient hypothesis is feasible in E .

Lean: `restrictFeasible_id`, `restrictFeasible_comp`, and `mem_range_restrictFeasible_iff`.

Thus $D \mapsto \mathcal{N}(D)$ is already a contravariant set-valued construction on the poset of active document sets. Calling it a presheaf becomes literal once the corpus-dependent universe and a target category are fixed. Calling it a sheaf would require a further gluing theorem; that is an open problem, not terminology assumed here.

The quotient moves the other way. Since

$$\text{Must}_D(c, d) \implies \text{Must}_E(c, d),$$

old must-classes map into new must-classes. Active-claim inclusion and relation preservation induce

$$q_{D,E} : F'_D \longrightarrow F'_E.$$

Lean: `includeActiveClaim` and `canonicalMap`. This map need not be injective: added evidence can eliminate all split worlds and thereby merge old must-classes. It also need not be surjective because E contains new active claims.

There is generally no quotient map in the deletion direction. A pair forced together in E may become uncertain in D after the supporting document is removed. Mapping one E -class back would then require splitting it, which is not a function on quotient classes. This is the formal reason that deletion cannot be implemented as the inverse of an agglomerative merge history.

7 Minimal document supports

An explanation should identify documents sufficient for a settled conclusion. For a finite document set D , say D supports a must-link when $\mathcal{N}(D)$ is nonempty and $\text{Must}_D(c, d)$ holds. Define cannot-support analogously. A support is minimal when no proper subset still supports the conclusion.

Theorem 7.1 (Finite minimal certificates). *If a finite corpus supports a must-coreference conclusion, it contains an inclusion-minimal finite support for that conclusion. The same holds for a cannot-coreference conclusion.*

Proof. Among the finite supporting subsets of the original corpus, choose one of minimum cardinality. Every proper subset has smaller cardinality and therefore cannot support the conclusion. $\square$

Lean: the generic finite minimization theorem is `exists_minimal_subfinset`; its two applications are `exists_minimal_mustSupport` and `exists_minimal_cannotSupport`.

The theorem is semantic, not an efficiency claim. Computing all minimal supports may be exponential. ATMS labels, unsatisfiable cores, provenance polynomials, and hitting-set dualization are candidate representations. The proof does establish a precise contract for an explanation engine: a returned certificate should be sufficient, consistent, and inclusion-minimal.

8 Lifting truth over normalization worlds

Let T be a type of downstream truth worlds and let

$$T : \Omega \longrightarrow \mathcal{P}(\mathsf{T})$$

assign a truth feasible set to each exact normalization. The joint feasible set is

$$J(D) = \{(\omega, t) \mid \omega \in \mathcal{N}(D), t \in T(\omega)\}.$$

This is `TruthLift.jointFeasible`. A property $P(\omega, t)$ is certain when it holds throughout $J(D)$ and possible when it holds somewhere in $J(D)$.

Theorem 8.1 (Truth-lift monotonicity). *If $D \subseteq E$, every property certain over $J(D)$ is certain over $J(E)$, and every property possible over $J(E)$ was possible over $J(D)$.*

Lean: `certain_mono` and `possible_antitone`.

Not every feasible normalization need support a downstream truth world. The projection of $J(D)$ onto normalization hypotheses is contained in $\mathcal{N}(D)$, with equality if every feasible normalization has a nonempty truth fiber. Lean: `normalizationProjection_subset` and `normalizationProjection_eq`.

The first commutation result covers the clean independence case.

Theorem 8.2 (Normalization-independent commutation). *Assume $\mathcal{N}(D) \neq \emptyset$, $T(\omega) = T_0$ for every ω , and $P(\omega, t) = P_0(t)$. Then*

$$\forall(\omega, t) \in J(D), P_0(t) \iff \forall t \in T_0, P_0(t).$$

Existentially, joint possibility is equivalent to consistency of $\mathcal{N}(D)$ together with ordinary possibility in T_0 .

Lean: `certain_constant_iff` and `possible_constant_iff`.

This identifies the exact modularity boundary. Normalization-first inference commutes automatically when truth semantics ignores normalization. The research problem is to weaken that assumption: characterize dependent $T(\omega)$ and $P(\omega, t)$ for which certain answers can be computed without enumerating every joint world.

9 Status evolution under corpus change

The modal laws give a concise transition discipline. Under $D \subseteq E$:

$$\begin{aligned} \text{Must}_D(c, d) &\implies \text{Must}_E(c, d), \\ \text{Cannot}_D(c, d) &\implies \text{Cannot}_E(c, d), \\ \text{May}_E(c, d) &\implies \text{May}_D(c, d), \\ \text{U}_E(c, d) &\implies \text{U}_D(c, d). \end{aligned}$$

Thus addition can resolve uncertainty into must or cannot, or make the entire problem inconsistent. It cannot turn a consistent settled fact into uncertainty within the hard-constraint semantics. Deletion reverses the information order: a settled link may become uncertain, but a possibility cannot be invented by adding a constraint.

These are `mustSame_mono`, `cannotSame_mono`, `maySame_antitone`, and `uncertain_antitone`. They are useful both as theorems and as property-based tests for an incremental implementation.

10 Relationship to learned coreference

Nothing in the theory forbids neural models. They can propose candidate frame interpretations, candidate links, and likely high-reduction criteria. They can rank feasible worlds. What they cannot do silently is turn an uncalibrated score threshold into a semantic impossibility.

This yields three distinct interfaces:

1. *generation*: place defensible worlds inside the ambient candidate universe;
2. *hard admissibility*: apply audited property sets by STE;
3. *ranking*: order the worlds that remain.

Candidate recall is a fairness condition on generation. Hard-constraint soundness is a fairness condition on property sets. Ranking does not alter must/cannot guarantees unless the ranked tail is explicitly discarded, in which case that discard is itself a property-set application that must be named and audited.

11 Mechanization map

The development is split by responsibility:

Lean module	Formal content
<code>Ste.DynamicFrame</code>	model, feasibility, modal relations, quotient
<code>Ste.DynamicFrame.Laws</code>	dynamic laws, restriction, status, quotient map
<code>Ste.DynamicFrame.Solver</code>	property application, reduction, finite runs
<code>Ste.DynamicFrame.Counterexample</code>	three-mention nontransitivity model
<code>Ste.DynamicFrame.Support</code>	finite minimal document certificates
<code>Ste.DynamicFrame.PresheafGluing</code>	partial restriction and exact gluing
<code>Ste.DynamicFrame.FiniteSolver</code>	executable solver and tight bounds
<code>Ste.DynamicFrame.RelationAlgebra</code>	audited typed necessary relations
<code>Ste.DynamicFrame.TruthLift</code>	dependent downstream feasible family
<code>Ste.DynamicFrame.Lineage</code>	occurrence-set identifiers and lineage
<code>Ste.DynamicFrame.AmbiguitySplit</code>	insertion-induced envelope splitting

The root module `Ste.lean` imports the complete theory. The project is pinned to Lean 4.32.0 and Mathlib 4.32.0. The repository workflow builds the root library on pushes to the development branch. The source and this paper use a strict convention: a named Lean theorem is the authoritative statement. The next section replaces the earlier conjectures by qualified theorems and records the boundary at which each claim becomes false.

12 Resolution of the six conjectures

The original conjectures mixed extensional existence claims with computational ones. None should be accepted without its domain assumptions. This section states what is actually achievable, proves it, and records sharp constants or small counterexamples where they matter.

12.1 Variable-universe restriction is necessarily partial

Write $W(D) = \mathcal{N}(D)$ for the subtype of exact worlds feasible at corpus D . For $D \subseteq E$, define

$$\rho_{E,D} : W(E) \longrightarrow W(D), \quad \rho_{E,D}(h) = h.$$

The earlier restriction proofs immediately give

$$\rho_{D,D} = \text{id}, \quad \rho_{E,D} \circ \rho_{F,E} = \rho_{F,D}.$$

Lean: `restrictWorld_id` and `restrictWorld_comp`. Thus feasible worlds form a contravariant functor on the document-set poset without any new axiom.

The active claim universe behaves differently. Insertion gives a total map $\mathcal{C}(D) \rightarrow \mathcal{C}(E)$, but deletion cannot give a total reverse map: a claim whose source lies in $E \setminus D$ has no active target. Define a partial restriction relation by equality of immutable occurrence identifiers.

Theorem 12.1 (Partial claim restriction). *For $D \subseteq E$ and $c \in \mathcal{C}(E)$, a restriction of c to $\mathcal{C}(D)$ exists exactly when the source document of c lies in D ; when it exists, it is unique.*

Proof. If a restricted active claim is equal to c , its source is active in D . Conversely, source membership constructs the subtype element directly. Equality of two such subtype elements follows from equality of their immutable occurrences. $\square$

Lean: `claimRestriction_exists_iff` and `claimRestriction_unique`. The exact fiber constant is therefore 0 or 1, not always 1. Local strict-coreference is preserved definitionally by restricting the world and including the surviving claims (`localSame_restrict`). This is the precise variable-universe result; a total deletion map was the false part of the unqualified conjecture.

12.2 Gluing is controlled by cross-document constraints

For an ambient hypothesis h , define the obstruction

$$\begin{aligned} \mathcal{O}_{D,E}(h) = \{k \in \mathbf{K} : \text{supp}(k) \subseteq D \cup E, \\ \text{supp}(k) \not\subseteq D, \text{supp}(k) \not\subseteq E, h \notin S_k\}. \end{aligned}$$

Only genuinely cross-document constraints occur in this set.

Theorem 12.2 (Exact union obstruction). *For every h ,*

$$h \in \mathcal{N}(D \cup E) \iff h \in \mathcal{N}(D) \wedge h \in \mathcal{N}(E) \wedge \mathcal{O}_{D,E}(h) = \emptyset.$$

Proof. The forward direction follows by feasibility antitonicity and satisfaction of every union-active constraint. Conversely, a union-active constraint is active on D , active on E , or active on neither side. The first two cases follow from local feasibility. In the third case, violation would place the constraint in $\mathcal{O}_{D,E}(h)$, contradicting emptiness. $\square$

Lean: `mem_feasible_union_iff`.

Two local views are called compatible here when they expose the same ambient exact hypothesis. Under that qualification, the global-extension type is nonempty exactly when the obstruction vanishes (`globalExtension_nonempty_iff`), and it is a subsingleton (`globalExtension_subsingleton`). Hence the number of extensions is at most 1. For finite $\mathbf{K}$ with decidable support and satisfaction, the obstruction is computable by one filter and contains at most $|\mathbf{K}|$ elements. If compatibility is weakened to agreement only on overlap, uniqueness is no longer promised: residual identity choices outside the overlap are precisely additional global worlds.

12.3 The finite exact solver and its tight limit

Assume the ambient exact hypothesis universe is finite of cardinality

$$N = |\Omega|,$$

and satisfaction is decidable. Represent a solver state by a finite set and implement property application by filtering.

Theorem 12.3 (Executable exactness). *Finite filtering denotes exactly abstract STE intersection, and a finite run denotes exactly the corresponding abstract run and partial feasibility set.*

Proof. Membership after filtering is membership in the input conjoined with satisfaction. Induction on the constraint list then identifies every finite run with repeated property-set intersection. $\square$

Lean: `coe_applyFinite`, `coe_runFinite`, and `coe_solveFinite`.

Every nontrivial application removes at least one world (`applyFinite_card_lt`). Total reduction from any state is at most N (`reductionCost_le_maxWorlds`). These are useful operational constants: the explicit representation stores at most N survivors, a full must/may scan examines at most N worlds, and no execution can eliminate more than N distinct worlds.

The unconditional compactness conjecture, however, is false.

Theorem 12.4 (Tight extensional state bound). *There are exactly 2^N extensional solver states over N ambient worlds, and unrestricted crisp STE can realize every one of them with a single property set.*

Proof. The solver states are the powerset of the N -element universe, whose cardinality is 2^N . Given any $A \subseteq \Omega$, use one constraint with property set $S_\star = A$; its feasibility set is exactly A . $\square$

Lean: `card_allExactStates` and `arbitrary_subset_as_single_property`. Therefore no uniform subexponential extensional representation follows from STE alone. BDDs, decomposable circuits, treewidth bounds, or factorized partition structures remain worthwhile, but their compactness theorems must assume structure in the generated property sets.

12.4 Typed relations lift exactly when their rules are audited

The mechanization gives each relation kind a source and target sort and keeps strict identity, subevent, supersession, before, causes, repeated instance, and framing divergence distinct. A concrete world semantics supplies the laws it actually validates: transitivity for subevent, supersession, and before; causation implying temporal precedence; equivalence laws for repeated instance; and symmetry plus irreflexivity for framing divergence.

Theorem 12.5 (Necessary-relation lifting). *Let a composition rule*

$$R_i(x, y) \wedge R_j(y, z) \implies R_k(x, z)$$

hold in every exact world. Replacing each relation by its must-version—the intersection over feasible worlds—preserves the same rule.

Proof. Fix a feasible world h . Both must-premises hold in h , so world-level soundness gives the conclusion in h . Since h was arbitrary, the must-conclusion follows. $\square$

Lean: `CompositionRule` and `must_compose`. Specialized proofs are `mustSubevent_trans`, `mustSupersession_trans`, `mustBefore_trans`, and `mustCausesBefore`; the repeated-instance and framing-divergence laws are lifted separately. The proof preserves one rule application exactly: two necessary premises produce one necessary conclusion. No theorem asserts, for example, transitivity of causation or framing divergence. Names do not license composition constants; audited world laws do.

Consistency is required to recover endpoint data from a necessary statement, because universal relations over an empty feasible set are vacuous. This is explicit in `mustRel_typed`, `mustStrictIdentity_iff`, and `not_mustDivergence_self`.

12.5 Dependent truth commutes through its marginal union

For dependent truth fibers $T(h)$ define

$$T_{\exists}(D) = \{t : \exists h \in \mathcal{N}(D), t \in T(h)\}.$$

The fibers may vary arbitrarily.

Theorem 12.6 (Marginal commutation). *For every truth-only predicate Q ,*

$$\begin{aligned} \forall h \in \mathcal{N}(D) \forall t \in T(h), Q(t) &\iff \forall t \in T_{\exists}(D), Q(t), \\ \exists h \in \mathcal{N}(D) \exists t \in T(h), Q(t) &\iff \exists t \in T_{\exists}(D), Q(t). \end{aligned}$$

Proof. Both equivalences follow by unpacking the existential witness in the definition of $T_{\exists}(D)$. No fiber constancy or nonemptiness premise is used. $\square$

Lean: `certain_marginal_iff` and `possible_marginal_iff`.

A genuinely dependent query $P(h, t)$ also commutes if, on jointly feasible support, there is a $Q(t)$ with $P(h, t) \leftrightarrow Q(t)$. This checkable condition is `SupportedInvariant`; the exact universal and existential results are `certain_supportedInvariant_iff` and `possible_supportedInvariant_iff`. The loss constant is zero: these are logical equivalences, not approximations. The condition is also meaningful. Two feasible normalizations sharing one truth world but disagreeing on P rule out every truth-only factor Q ; this two-normalization, one-truth-world witness is formalized by `no_supportedInvariant_of_disagreement`.

12.6 Stable identity is a member set; history is a relation

For a canonical must-class q at snapshot D , define

$$\text{id}_D(q) = \{c \in \mathcal{C}(D) : c \text{ belongs to } q\}.$$

These are immutable claim occurrence identifiers, not generated cluster names.

Theorem 12.7 (Complete snapshot identifier). *Every $\text{id}_D(q)$ is nonempty, and $\text{id}_D(q) = \text{id}_D(r)$ implies $q = r$. For $D \subseteq E$,*

$$\text{id}_D(q) \subseteq \text{id}_E(q_{D,E}(q)).$$

Proof. A representative belongs to its own equivalence class. Equal member sets put each representative in the other's class, so quotient soundness gives equal classes. Under insertion, claims remain active and must-coreference is monotone, proving the inclusion. $\square$

Lean: `canonicalMembers_nonempty`, `canonicalMembers_injective`, and `canonicalMembers_subset_map`. Thus the minimum identifier size is 1, and the identifier has zero collisions inside a snapshot.

Across arbitrary snapshots define lineage by nonempty intersection of member sets. It is reflexive for live frames, symmetric, and every frame has lineage to its insertion image (`lineage_refl`, `lineage_symm`, and `lineage_canonicalMap`). It is intentionally not transitive: the sets $\{a\}$, $\{a, b\}$, and $\{b\}$ give the smallest counterexample. Lean proves this on a two-element type in `overlap_not_transitive`. Consequently a single immutable scalar identifier cannot simultaneously model a merge and both later split children without collision or arbitrary choice. Stable snapshot identity plus relational lineage is achievable; globally functional identity through splits is not.

12.7 Achievability constants

Result	Constant	Qualification or sharp limit
Claim restriction	fiber size 0 or 1	source must remain active
Compatible gluing	extensions ≤ 1	locals name the same ambient world
Finite exact solver	N survivors; 2^N states	exponential bound is tight
Typed composition	one sound rule step	only audited rules lift
Truth commutation	zero logical loss	invariant on feasible support
Snapshot identity	size ≥ 1 ; zero collisions	history is overlap, not a function
Ambiguity-envelope split	reduction ≥ 1 ; at most $ C $	old may-link eliminated by evidence

Other engineering questions retain exact correctness oracles: incremental output must equal a fresh STE meet; insertion order must not matter; deletion must reactivate every world excluded only by removed support; and every published settled link should admit a finite minimal certificate when the active corpus is finite.

13 Conclusion

Dynamic frame normalization is best understood as set-theoretic estimation over exact partitions, not as an irreversible clustering task. Property-set application is the solver. Provenance selects the active meet. Must coreference is the intersection of feasible equivalence relations and supplies the only generally safe deterministic quotient. ‘May’ and $\mathbf{U}$ are modal summaries, and a finite machine-checked countermodel shows why they cannot be collapsed into partitions. Nevertheless, a database-facing ambiguity envelope can genuinely split under insertion: new properties remove merge worlds, shrink `May`, and can force formerly compatible descriptions apart. The Toyota/BMW witness makes this distinction constructive. Document addition and deletion act in opposite information directions; feasible worlds restrict contravariantly, while canonical quotients map covariantly under addition. Minimal finite supports provide an explanation contract, and the dependent truth lift keeps normalization uncertainty intact downstream.

The result is a substantive mathematical kernel for a live document database: ambiguity is represented, reduction is exact, retraction has a denotation, cross-document gluing has an obstruction, and finite implementation has both an executable correctness theorem and a tight worst-case bound. The earlier open conjectures are no longer slogans: each is either proved under explicit hypotheses or delimited by a machine-checked counterexample. The principal remaining algorithmic question is structural rather than semantic—which properties of real candidate graphs make

their exact 2^N worst case compress in practice while retaining deletion, explanation, and must/may queries.

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